

AutoInsight: Digital Enquiry and BI Reporting System

Phase 2 Logical Data Relationship Diagram

Client	UK Auto Parts SME
Phase	Phase 2 - Future Enhancement
Prepared by	Business Analyst
Document date	June 20XX
Purpose	Define the logical data relationships required for customer enquiries, quote requests, lead tracking, follow-up activity, newsletter consent and future Power BI reporting.

1. Document Purpose

This document defines the business-level data entities and relationships required before final SQL database design and Power BI semantic model implementation. It supports requirement validation, reporting preparation and technical discussion.

Important distinction: The logical entities in this document are not a copy of the current Power BI Model view. The current Power BI file uses denormalised SQL reporting views beginning with vw_. This document describes the underlying business information and intended relationships.

2. Scope and Design Status

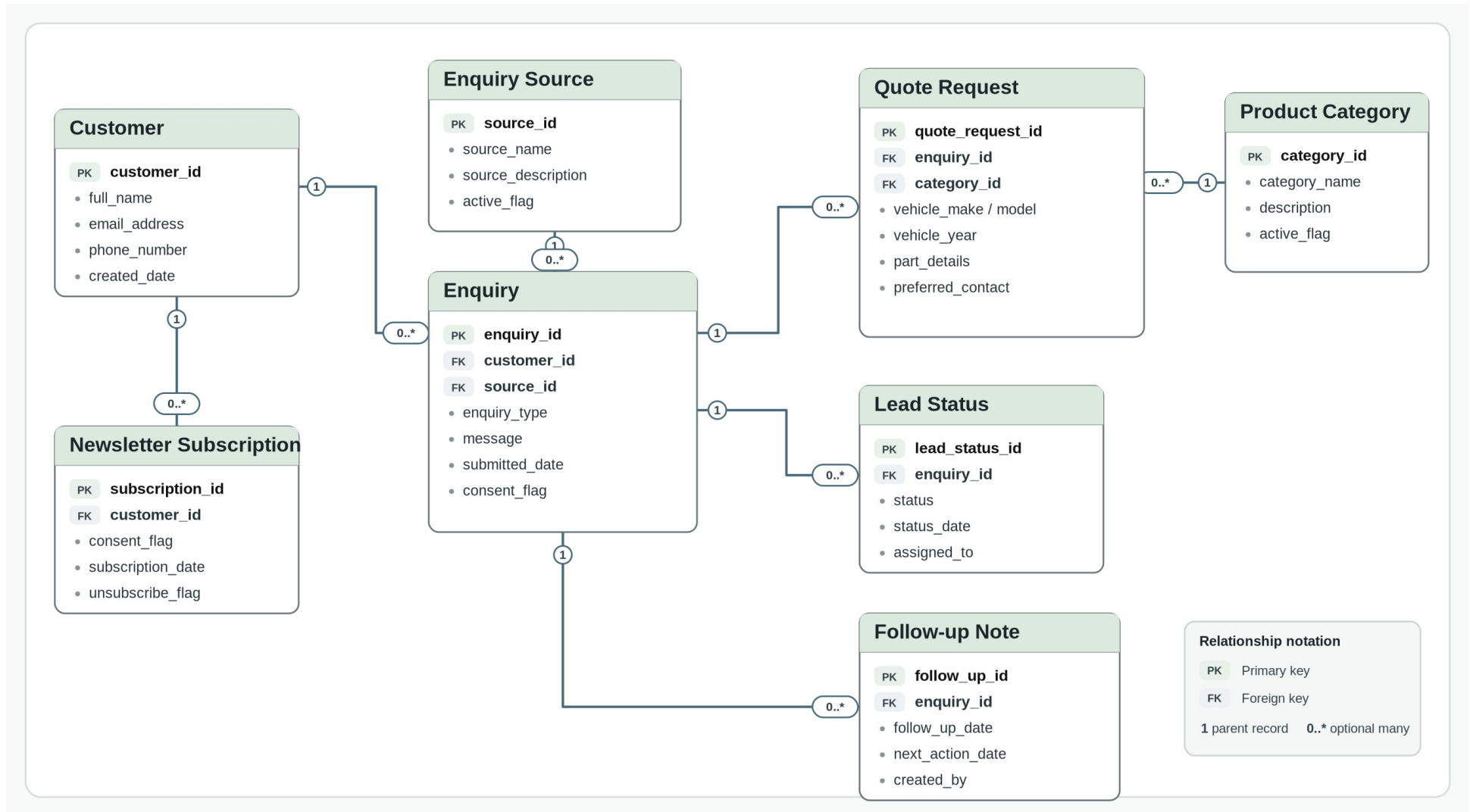
Logical scope: Customer, Enquiry, Enquiry Source, Quote Request, Product Category, Lead Status, Follow-up Note and Newsletter Subscription.

Reporting scope: Enquiry volumes, quote activity, lead progress, follow-up workload, enquiry source, product category, conversion rate and monthly trends.

Technical status: This is not the final physical database design. Keys, data types, indexes, security controls, retention rules and integration logic must be confirmed by the technical team and relevant business or compliance owner.

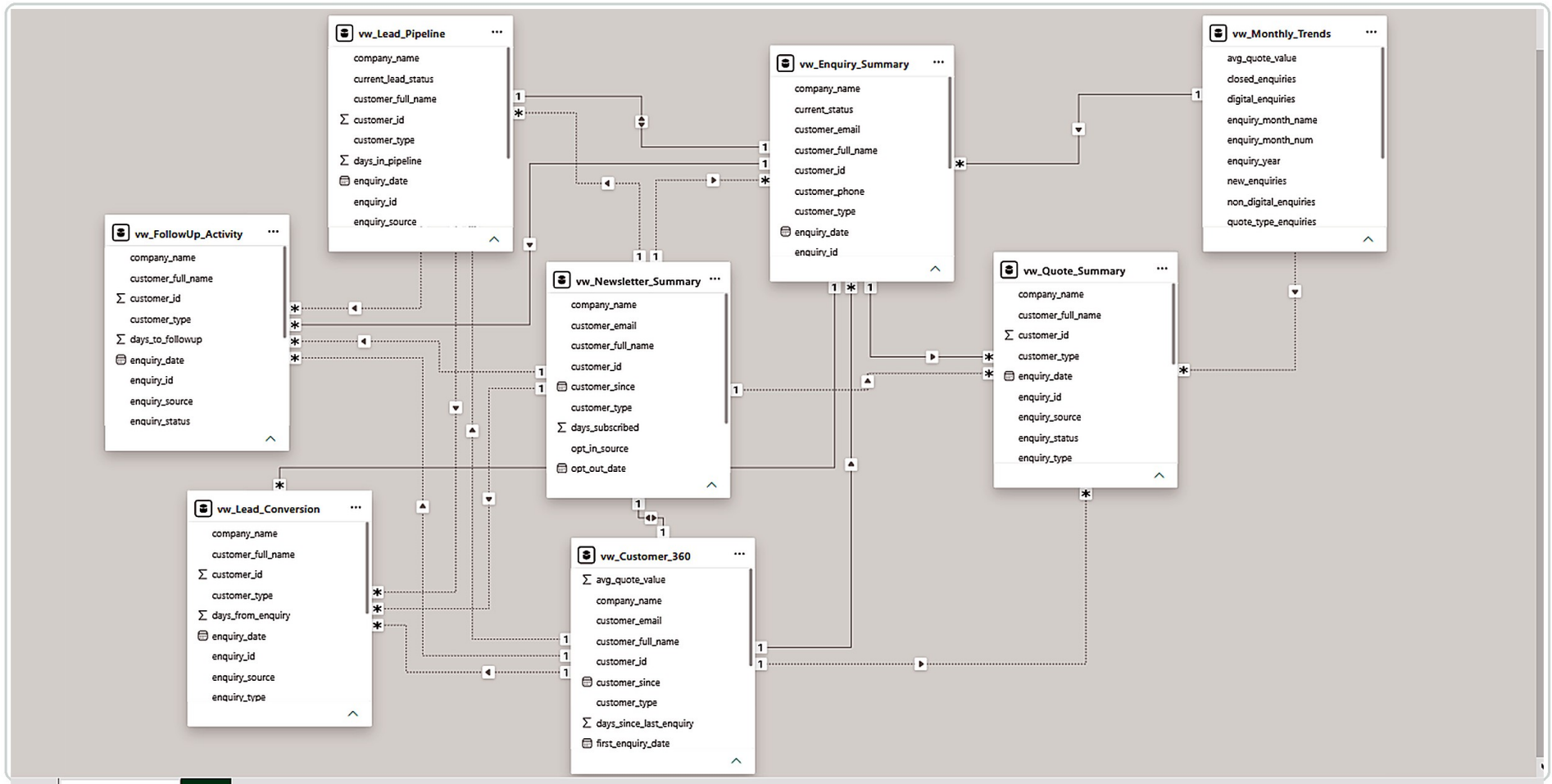
3. Logical Data Relationship View

Simplified business-level model. Each child relationship is optional and can contain zero or many records.



4. Implemented Power BI Model View

Cropped from the supplied Power BI Desktop screenshot. The model contains eight SQL reporting views and includes active and inactive view-to-view relationships.



Purpose: Portfolio evidence of the reporting model implemented in Power BI.

Interpretation: These are reporting views, not a one-to-one physical database schema.

5. Logical Entities and Relationship Summary

Logical Entity	Business Purpose
Customer	Stores customer contact details captured through enquiry, quote, feedback or newsletter forms.
Enquiry	Stores the main website enquiry record and its source, type, message, date and consent indicator.
Enquiry Source	Stores standard enquiry origins such as quote form, contact form, feedback form or newsletter.
Quote Request	Stores quote-specific details including requested category, vehicle details, part description and contact preference.
Product Category	Stores approved auto parts categories used by the website and reporting layer.
Lead Status	Stores status history such as New, Contacted, Quoted, Won, Lost or Closed.
Follow-up Note	Stores follow-up notes, owner, follow-up date and next action date for an enquiry.
Newsletter Subscription	Stores newsletter consent, subscription date and unsubscribe status history.

Relationship	Meaning
Customer to Enquiry	One customer can submit zero or many enquiries. Each enquiry belongs to one customer where customer details are captured.
Customer to Newsletter Subscription	One customer can have zero or many subscription history records to retain consent and unsubscribe changes.
Enquiry Source to Enquiry	One enquiry source can be linked to zero or many enquiries for source reporting.
Enquiry to Quote Request	One enquiry can create zero or many quote requests when a customer asks for pricing or availability.
Product Category to Quote Request	One product category can be selected in zero or many quote requests.
Enquiry to Lead Status	One enquiry can have zero or many status updates throughout follow-up.
Enquiry to Follow-up Note	One enquiry can have zero or many follow-up notes and next actions.

6. Mapping to the Current Power BI Reporting Model

The supplied Power BI Model view contains eight SQL reporting views. These views combine and summarise the logical entities for report use. The mapping below prevents the logical data design from being confused with the Power BI semantic layer.

Logical item	Power BI view or field	Interpretation
Customer	vw_Customer_360	Customer-level reporting view. Customer fields also repeat in other views.
Enquiry	vw_Enquiry_Summary	Main enquiry reporting view.
Quote Request	vw_Quote_Summary	Quote and enquiry details prepared for quote reporting.
Follow-up Note	vw_FollowUp_Activity	Follow-up workload and activity reporting.
Lead Status	vw_Lead_Pipeline; vw_Lead_Conversion	Lead progress, current status and conversion reporting.
Newsletter Subscription	vw_Newsletter_Summary	Subscription, opt-in and opt-out reporting.
Enquiry Source	Columns inside reporting views	Visible as enquiry_source or customer_source rather than a separate Power BI dimension.
Product Category	Columns inside reporting views	Visible as category or enquiry-type attributes rather than a separate Power BI dimension.
Monthly reporting aggregate	vw_Monthly_Trends	Monthly enquiry and quote measures. This is a reporting aggregate, not a logical transaction entity.

7. BA Validation Notes

Validation Area	BA Note
Data completeness	Customer name and email should be captured for quote and contact enquiries where required by the form.
Consent	Consent should be captured where personal data is used to respond or send marketing communication.
Status standardisation	Lead status values should use one approved list to prevent Power BI reporting mismatches.
Category standardisation	Product category values should match the approved website categories and SQL transformation rules.
Follow-up tracking	Follow-up activity should include follow-up date, next action date and owner where required.
Reporting readiness	Data should support total enquiries, quote requests, top categories, pending follow-ups, conversion rate and monthly trend.
Power BI relationship control	The screenshot contains active and inactive view-to-view relationships. Exact relationship columns, cardinality and filter direction should be documented separately in a Power BI Model Specification.